Elevated Serum Pesticide Levels and Risk of Parkinson Disease

Background Exposure to pesticides has been reported to increase the risk of Parkinson disease (PD), but identification of the specific pesticides is lacking. Three studies have found elevated levels of organochlorine pesticides in postmortem PD brains. Objective To determine whether elevated levels of organochlorine pesticides are present in the serum of patients with PD. Design Case-control study. Setting An academic medical center. Participants Fifty patients with PD, 43 controls, and 20 patients with Alzheimer disease. Main Outcome Measures Levels of 16 organochlorine pesticides in serum samples. Results β-Hexachlorocyclohexane (β-HCH) was more often detectable in patients with PD (76%) compared with controls (40%) and patients with Alzheimer disease (30%). The median level of β-HCH was higher in patients with PD compared with controls and patients with Alzheimer disease. There were no marked differences in detection between controls and patients with PD concerning any of the other 15 organochlorine pesticides. Finally, we observed a significant odds ratio for the presence of β-HCH in serum to predict a diagnosis of PD vs control (odds ratio, 4.39; 95% confidence interval, 1.67–11.6) and PD vs Alzheimer disease (odds ratio, 5.20), which provides further evidence for the apparent association between serum β-HCH and PD. Conclusions These data suggest that β-HCH is associated with a diagnosis of PD. Further research is warranted regarding the potential role of β-HCH as a etiologic agent for some cases of PD.